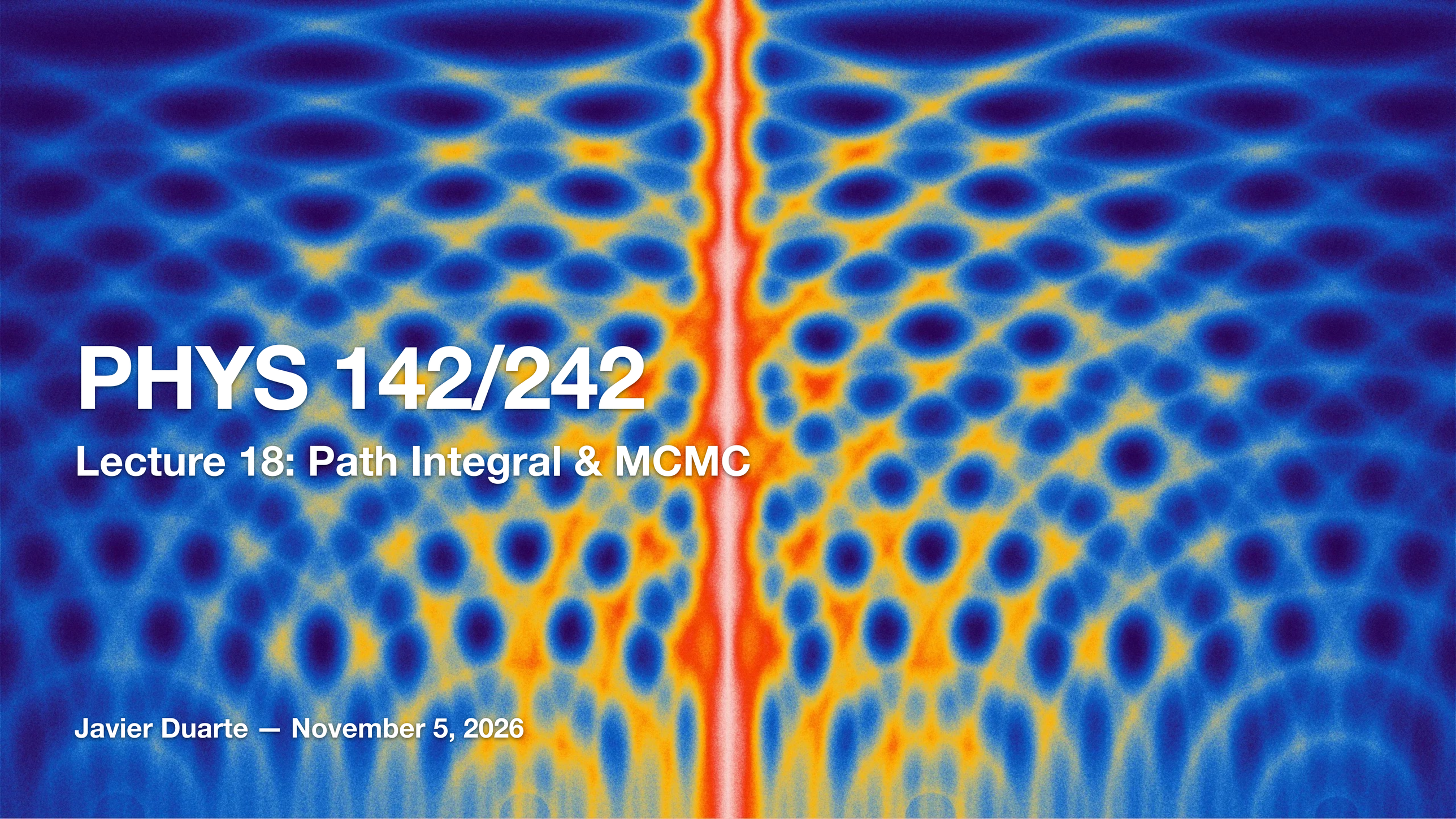


PHYS 142/242

Lecture 18: Path Integral & MCMC

Javier Duarte — November 5, 2026

Path integral

We've studied the path integral to evaluate the propagator, which represents the probability amplitude of a particle at spacetime point (x_a, t_a) reaching spacetime point (x_b, t_b)

$$K(x_b, t_b; x_a, t_a) = \int \mathscr{D}x(t) \exp\left[\frac{i}{\hbar}\int_{t_a}^{t_b} L(x(t))dt\right]$$

where $L(x(t)) = \frac{m}{2}\left(\frac{dx}{dt}\right)^2 - V(x)$ is the classical Lagrangian, $\mathscr{D}x(t)$ denotes the integral over all spacetime paths between (x_a, t_a) and (x_b, t_b)

Imaginary-time path integral

An alternative formulation uses imaginary time, where t is replaced by $-i\tau$

$$\tilde{\rho}(x_b, \tau_b; x_a, \tau_a) = \int \mathscr{D}x(\tau) \exp\left[-\frac{1}{\hbar}\int_{\tau_a}^{\tau_b} L_E(x(\tau))d\tau\right]$$

where $L_E(x(\tau)) = \frac{m}{2}\left(\frac{dx}{d\tau}\right)^2 + V(x(\tau))$ is the "Euclidean Lagrangian"

Just as for real times, the path integral is over all paths between (x_a, τ_a) and (x_b, τ_b)

Connection to quantum stat. mech.

If we set $x_b = x_a = x$ and integrate over x (i.e., take the trace), $\tau_a = 0$, and $\tau_b = \hbar\beta$ where $\beta = 1/(k_{\mathrm{B}}T)$, we get partition function

$$Z = \int \mathscr{D}x(\tau) \exp\left[-\frac{1}{\hbar}\oint_0^{\hbar\beta} L_E(x(\tau))d\tau\right]$$

which alternatively can be calculated from the Boltzmann distribution

$$Z = \sum_{n=0}^{\infty} e^{-\beta E_n}$$

Calculating the ground state

Note, as we approach $T \to 0$, i.e. $\tau_b = \hbar\beta \to \infty$ (more precisely, $\beta(E_1 - E_0) \gg 1$)

$Z = \sum_{n=0}^{\infty} e^{-\beta E_n} \sim e^{-\beta E_0}$ (i.e. only the first eigenvalue dominates the sum)

and

$$\rho(x, \hbar\beta; x, 0) = \sum_{n}^{\infty} p_n |\phi_n(x)|^2 \sim |\phi_0(x)|^2 \quad \text{where} \quad p_n = e^{-\beta E_n}/Z$$

So if we run with a long enough τ_b, our probability distribution should converge to the ground state.

We can use the path integral (with imaginary time) to directly calculate the ground state!

Imaginary-time path

Discretize “time” with N increments, $\tau_a = 0$, and $\tau_b = N\eta$

Each vector $(x_0, x_1, \ldots, x_{N-1})$, where $x_N \equiv x_0$, in N-dimensional space represents an imaginary-time path

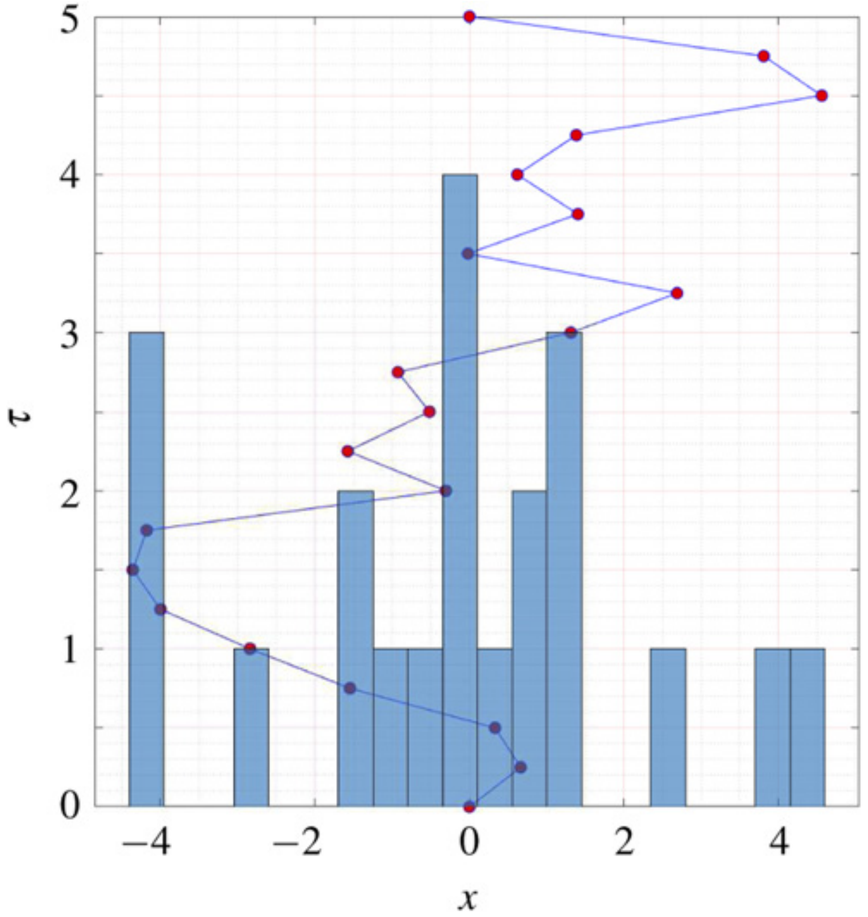
The probability density over this N-dimensional space can be written in terms of the discretized ‘Euclidean’ action S_E

$$S_E = \sum_{i=0}^{N-1} \left[\frac{m\,(x_{i+1} - x_i)^2}{2\eta} + \eta\, V(x_i) \right] \approx \oint_0^{\hbar\beta} L_E\big(x(\tau)\big)\, d\tau$$

Then,

$$p(x_0, \ldots, x_{N-1}) = \frac{1}{Z} \left(\frac{2\pi\hbar\eta}{m} \right)^{-N/2} e^{-S_E/\hbar}$$

Calculating the ground state



S. Mittal et al., “Path integral Monte Carlo method for the quantum anharmonic oscillator”, Eur. J. Phys. 41 055401 (2020)

A discrete imaginary time path ⇒ histogram of the probability density by summing up the number of times a particle is in a bin

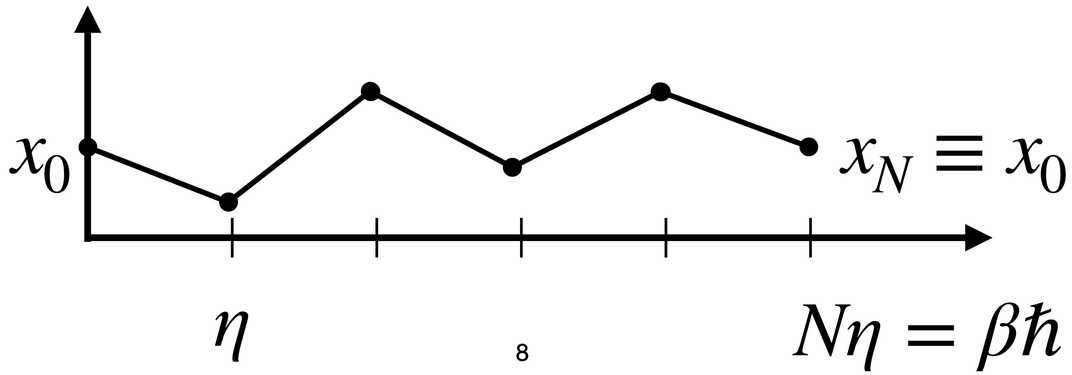
Figure 1. A discrete imaginary-time path between $(0, 0)$ and $(0, 5)$. The histogram indicates the number of times the particle crosses the corresponding spatial region.

Markov chain Monte Carlo method

Markov chain Monte Carlo simulations are carried out in discrete “time” with N increments, $\tau_a = 0$, and $\tau_b = N\eta$

$$Z = \lim_{\eta \to 0} \int \cdots \int dx_0 \cdots dx_{N-1} \left(\frac{2\pi\hbar\eta}{m} \right)^{-N/2} e^{-S_E/\hbar}$$

Note: $\eta \to 0$ with $N\eta = \hbar\beta$ held fixed



MCMC steps

Begin with an initial path, which may be an array of random numbers (‘hot’ start) or zeros (‘cold’ start).

Update the path by applying Metropolis algorithm to each element x_i of the path in random order (called a ‘sweep’)

(a) Generate a uniform random number $u \in [-h, h]$ (where h is the ‘hit size’)

(b) Propose the new value $x_i' = x_i + u$ of the path element and calculate the resulting change in the action: $\Delta S_E = \frac{m}{\eta}(x_i' - x_i)(x_i' + x_i - x_{i-1} - x_{i+1}) + \eta\,[V(x_i') - V(x_i)]$

(indices wrap around the ring $x_{-1} \equiv x_{N-1}; x_N \equiv x_0$)

If $\Delta S_E \leq 0$, accept the new path element.

If $\Delta S_E > 0$, accept with probability $\exp(-\Delta S_E/\hbar)$

MCMC considerations

One sweep produces the next path from the previous one

Each path is determined only by the immediately preceding path, so the complete sequence of paths forms a Markov chain, but the paths are correlated

The initial path ‘thermalizes’ (burn-in), that is, attains equilibrium after N_{therm} sweeps

To determine the (larger) error due to the inherent autocorrelation of the Markov chain, keep every sweep after burn-in, and get the error from τ_{int} or blocking

Example: Anharmonic oscillator

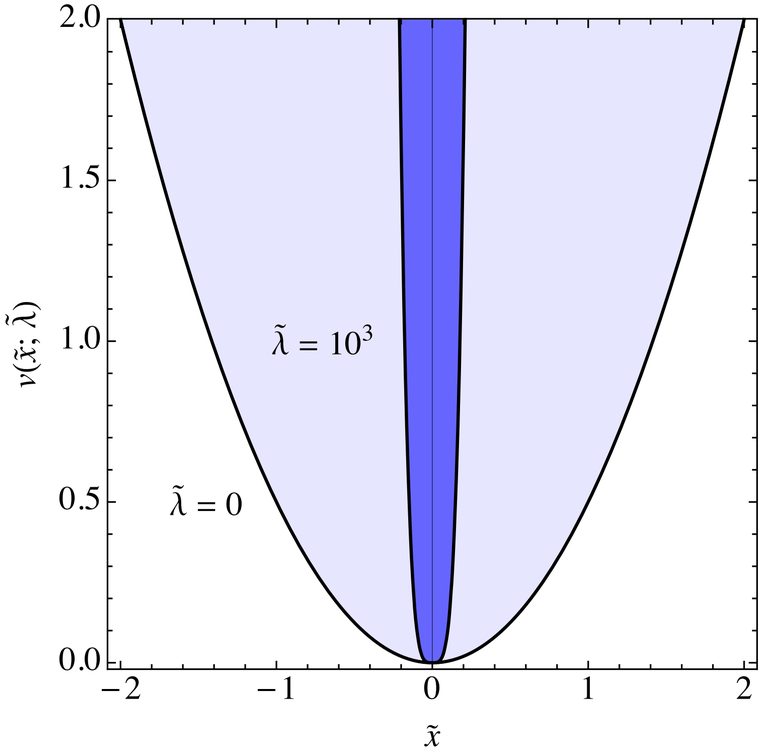


Figure 2. The potential $v(x;\lambda) = \frac{1}{2}\tilde{x}^2 + \tilde{\lambda}\tilde{x}^4$ for $\tilde{\lambda} = 0$, which is the potential for the harmonic oscillator (blue shading), and $\tilde{\lambda} = 10^3$, which corresponds to the strong quartic limit (blue shading). The quartic potential localizes the wave functions of the oscillators, which causes the corresponding energy eigenvalues to increase.

S. Mittal et al., Eur. J. Phys. 41 055401 (2020)

Consider the anharmonic oscillator:

$$\hat{H} = \frac{\hat{p}^2}{2m} + \frac{1}{2}m\omega^2\hat{x}^2 + \lambda\hat{x}^4$$

Or in “dimensionless” units (equivalent to setting $m = \omega = \hbar = 1$):

$$\hat{H} = \frac{\hat{p}^2}{2} + \frac{1}{2}\hat{x}^2 + \tilde{\lambda}\hat{x}^4$$

The discretized action is ($x_N \equiv x_0$):

$$S_E = \sum_{i=0}^{N-1}\left[\frac{(x_{i+1}-x_i)^2}{2\eta} + \eta\left(\frac{1}{2}x_i^2 + \tilde{\lambda}x_i^4\right)\right]$$

Probability density calculation

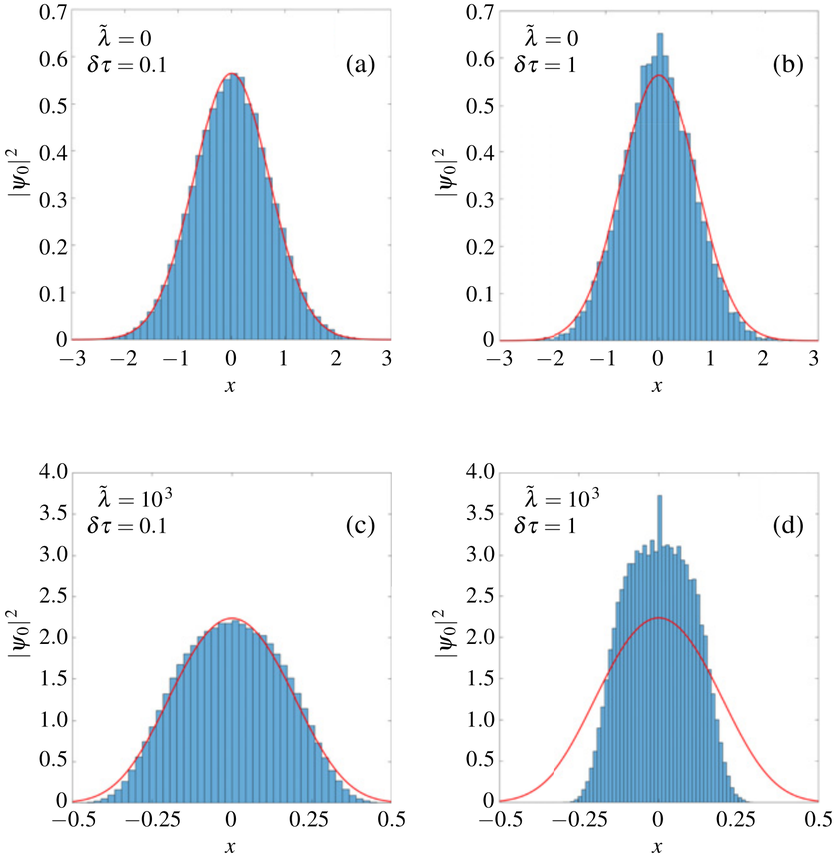


Figure 5. The probability densities $|\psi_0(x)|^2$ of the ground-state wave functions for the harmonic oscillator (a), (b) and the strong quartic limit (c), (d) in the Schrödinger equation with Hamiltonian (32) for $\delta\tau = 1$ (b), (d) and $\delta\tau = 0.1$ (a), (c). The histograms were obtained according to the procedure shown in figure 1, with 200 paths (every 100th from a chain of 20 000) used for determining the probability density. The red curves superimposed on these histograms are numerical solutions to Schrödinger's equations with the corresponding Hamiltonians (32) obtained by using the `bvp4c` solver of MATLAB (reference [44, 45]).[4]

S. Mittal et al., Eur. J. Phys. 41 055401 (2020)

Using histogramming over paths to find the ground state distribution

Better approximation to the ground state for smaller time step ($\eta = \delta\tau = 0.1$ vs. $\eta = \delta\tau = 1$)

Ground state energy calculation

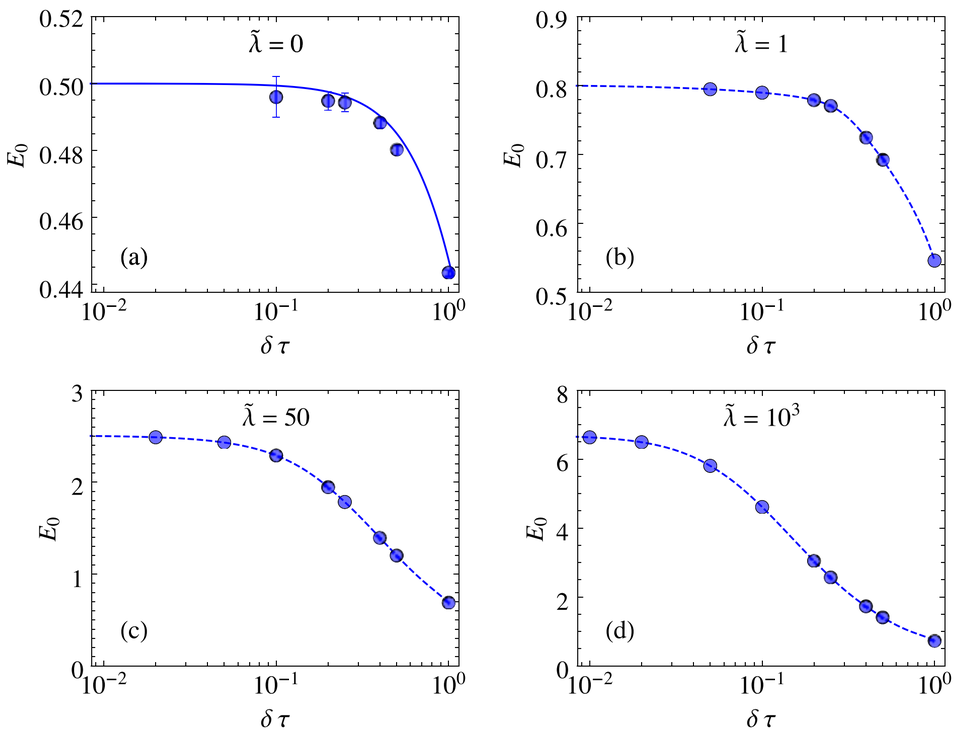


Figure 6. Calculation of the ground-state energy of the anharmonic oscillator with quartic couplings (a) $\tilde{\lambda} = 0$, (b) $\tilde{\lambda} = 1$, (c) $\tilde{\lambda} = 50$, and (d) $\tilde{\lambda} = 10^3$. The filled circles represent values calculated from the MCMC method. In (a), the solid line is the exact result calculated in reference [24, 30], while in (b)–(d), the broken curve is a (not-a-knot) cubic spline fit carried out on linear axes. The logarithmic axis for $\delta\tau$ is for presentation purposes only. Where error bars are not indicated, the errors are of the same size or smaller than the symbol.

S. Mittal et al., Eur. J. Phys. 41 055401 (2020)

Estimate E_0 using quantum virial theorem

$$E = \langle \hat{V} \rangle + \langle \hat{K} \rangle = \langle \hat{V} \rangle + \frac{1}{2}\langle \hat{x}\frac{d\hat{V}}{dx} \rangle$$
$$= \langle \hat{x}^2 \rangle + 3\tilde{\lambda}\langle \hat{x}^4 \rangle$$

Better approximation to the ground state for smaller time step ($\eta = \delta\tau = 0.1$ vs. $\eta = \delta\tau = 1$)

Beyond the ground state: finite β

At any β, the chain samples the thermal distribution

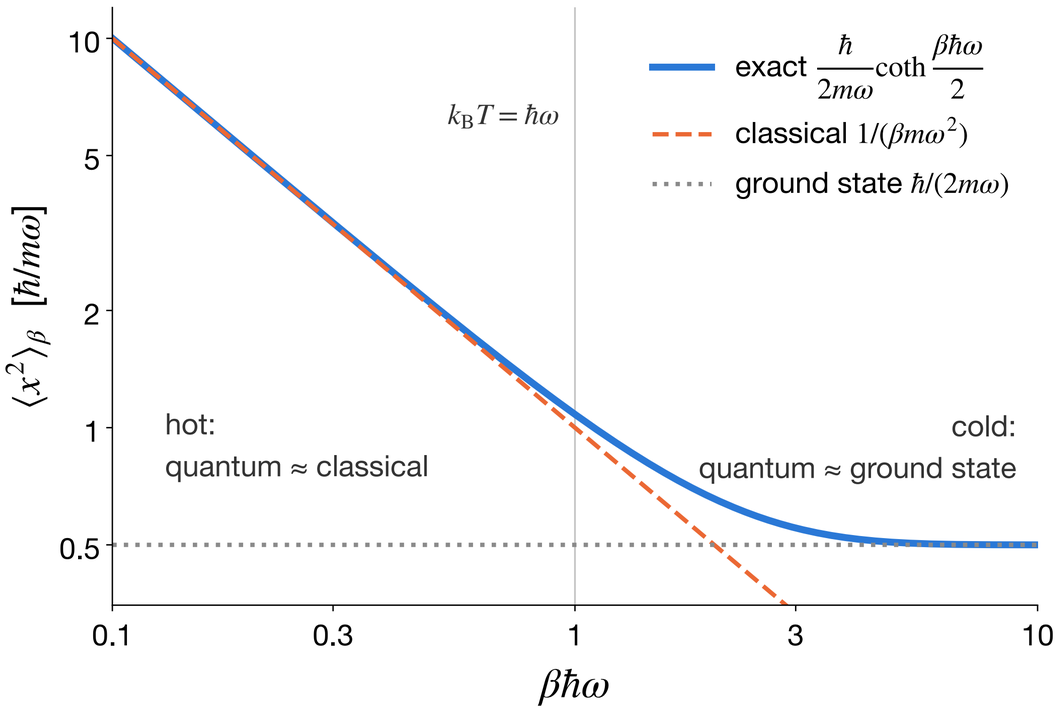
$$\frac{\rho(x,x)}{Z} = \sum_n p_n \,|\,\phi_n(x)\,|^2, \qquad p_n = \frac{e^{-\beta E_n}}{Z}$$

For harmonic oscillator, estimator is

$$\langle x^2 \rangle_\beta \approx \left\langle \frac{1}{N} \sum_{i=0}^{N-1} x_i^2 \right\rangle_{\text{sweeps}}$$

Two relevant limits:

$$\langle x^2 \rangle_\beta = \frac{\hbar}{2m\omega} \coth \frac{\beta\hbar\omega}{2} \;\to\; \begin{cases} \hbar/(2m\omega) & \text{ground state, } \beta\hbar\omega \gg 1 \\ 1/(\beta m\omega^2) & \text{classical equipartition, } \beta\hbar\omega \ll 1 \end{cases}$$



Recap: MCMC for Ising vs. Path integral

Ising	Path integral
lattice site	time slice
spin σ_i	position x_i
coupling J	spring m/η
field h	potential V
E/k_BT	$S_E/\hbar$
periodic lattice	trace
2D lattice	ring of N slices of length $\hbar\beta$